

**Q.1. Coin Tossing**

Through the simulation, show that probability of getting HEAD by tossing a fair coin is about 0.5. Write your observation from the simulation run.

**Q.2. Performance analysis of Bubble Sort**

Write the program to implement two different versions of bubble sort( BUBBLE SORT that terminates if the array is sorted before  $n-1^{\text{th}}$  Pass. Vs. BUBBLE SORT that always completes the  $n-1^{\text{th}}$  Pass) for randomized data sequence.

**Q.3. Variants of QUICK SORT**

Compare the performance of Quick Sort and Insertion Sort for instance characteristic  $n=10, \dots, 1000$ . Find out the cross-over point where Insertion Sort shows the better performance over Quick Sort. Modify your sorting algorithm to stop partitioning the list in Quick Sort when the size of the (sub) list is less than or equal to 12 (say) and sort the remaining sub list using Insertion Sort. Your counter will now have to count compares in both the partition function and every iteration of Insertion Sort. Again, run the experiment for 50 iterations and record the same set of statistics. Compare your results for the two different sorting techniques and comment upon your results.

***Upload your report [Source Codes + Simulation results +Observations] as a single pdf file with name as Roll No APL I to the specific assignment in MS Team***